

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

Strong password policies and MFA — prevents password sharing, weak passwords, and unauthorized access.

Firewall rules and port filtering — controls incoming and outgoing network traffic and blocks unnecessary connections.

Network access privileges — ensures employees only have access to the systems and data they need.

Part 2: Explain your recommendations

1. Strong password policies and MFA

The organization should require employees to use strong, unique passwords and should not allow password sharing. MFA should also be enabled, especially for administrator and other sensitive accounts. These measures reduce the risk of attackers gaining access using stolen or guessed passwords. Password policies should be enforced continuously, while account and authentication settings should be reviewed regularly.

2. Firewall rules and port filtering

The organization should configure firewall rules to allow only necessary network traffic and block unauthorized or unnecessary connections. Unused ports should also be blocked. This reduces the network's attack surface and makes it harder for attackers to access internal systems. Firewall rules should be reviewed and updated regularly as network requirements and threats change.

3. Network access privileges

The organization should limit users' access to only the systems and information

required for their jobs. This follows the principle of least privilege. If an employee's account is compromised, limited permissions can prevent the attacker from accessing the entire network or sensitive customer data. Access privileges should be reviewed regularly and removed when employees no longer need them.